

⌵ Hide menu

Simpson's Paradox, Weighting, & CMH

Case Control Sampling and Exact Inference for OR

2x2 Tables

Sign, Sign Rank, and Rank Sum

Poisson

Review

Practice Quiz: Module 4

Homework

Started

Quiz: Module 4 Quiz

Submitted

🎉 Congratulations! You passed!

Grade received 100%

Latest Submission

To pass 80% or higher

Go to next item

Module 4 Quiz

coach

1. A case control study of esophageal cancer was conducted. The data was stratified by 2 age groups. = low). The data was stratified by 2 age groups. 1 / 1 point

Low age group

H L

Case 8 5

Control 52 164

📌 Submit your assignment

Due Aug 3, 11:59 PM IST Attempts 3 every 8 hours Try again

High age group

H L

Case 23 21

Control 28 128

📌 Receive grade

To Pass 80% or higher

Your grade 100%

View Feedback

Assuming a constant odds ratio across age-strata, test to see if the odds ratio is 1 and report a P-value

👉 Correct

👍 Like

🗨️ Discuss

🚩 Report an issue

2. The Biostatistics and Epidemiology departments are running a 10K road race. There are three pairs of runners. In each case, a runner from Biostat was matched to a runner from Epi of the same age, gender and degree of running experience. The difference in each pairs times was taken and a signed rank test performed. 1 / 1 point

What is the smallest value that the two sided exact signed rank p-value could take?

👉 Correct

3. Two computer based methods for diagnostic imaging are being studied. Ten images were processed with both methods, resulting in a + or - diagnosis for each method and image. The data are as follows 1 / 1 point

+ -

+ 55 41

- 12 20

Where Method A is represented with the two columns and Method B is represented on the rows. What is the P-value for a of the hypothesis that the marginal probability of a positive diagnosis is the same for the two methods? (Use the large sample version.)

👉 Correct

4. Two computer based methods for diagnostic imaging are being studied. Ten images were processed with both methods, resulting in a + or - diagnosis for each method and image. The data are as follows 1 / 1 point

+ -

+ 55 41

- 12 20

Where Method A is represented with the two columns and Method B is represented on the rows. What is the marginal odds ratio of a positive diagnosis comparing the two methods? (Put the odds associated with Method B in the numerator.)

👉 Correct

5. A matched case control study was executed to study an airborne environmental toxicant's effect on lung cancer. The data are 1 / 1 point

Case and exposure status count

Case exposed, control exposed 243

Case exposed, control unexposed 189

Case unexposed, control exposed 54

Case unexposed, control unexposed 153

What is the **conditional** odds ratio of odds of exposure for cases over odds of exposure for controls?

👉 Correct

6. A topical rash treatment was applied to a portion of a rash on n patients. A quantitative measure of redness was calculated for the treated and untreated regions of the rash. A sign of + was given when the treated area was less red (more improved) than the untreated area and a - sign when it was not. It is desired to know whether the treatment improves the rash. 1 / 1 point

How many possible values can the P-value for the sign test take?

👉 Correct

7. A topical rash treatment was applied to a portion of a rash on n patients. A quantitative measure of redness was calculated for the treated and untreated regions of the rash. A sign of + was given when the treated area was less red (more improved) than the untreated area and a - sign when it was not. It is desired to know whether the treatment improves the rash. 1 / 1 point

Consider a 5% one sided sign test for 5 subjects. What would be the power of the test if the probability that the treatment works is actually 80% instead of the 50% assumed under the null?

👉 Correct

https://www.coursera.org/learn/biostatistics-2/exam/3rc7v/module-4-quiz/attempt/redirectToCoverPage

1/1